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Computational prediction of phytochemical inhibitors against the cap-binding domain of Rift Valley fever virus

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Abstract

Rift Valley fever is a zoonotic disease that can spread through livestock and mosquitoes, and its symptoms include retinitis, photophobia, hemorrhagic fever and neurological effects. The World Health Organization has identified Rift Valley fever as one of the viral infections that has potential to cause a future epidemic. Hence, efforts are urgently needed toward development of therapeutics and vaccine against this infectious disease. Notably, the causative virus namely, the Rift Valley fever virus (RVFV), utilizes the cap-snatching mechanism for viral transcription, rendering its cap-binding domain (CBD) as an effective antiviral target. To date, there are no published studies towards

identification of potential small molecule inhibitors for the CBD of RVFV. Here, we employ a virtual screening workflow comprising of molecular docking and molecular dynamics (MD) simulation, to identify 5 potential phytochemical inhibitors of the CBD of RVFV. These 5 phytochemical inhibitors can be sourced from Indian medicinal plants, *Ferula assa-foetida*, *Glycyrrhiza glabra* and *Leucas cephalotes*, used in traditional medicine. In sum, the 5 phytochemical inhibitors of the CBD of RVFV identified by this purely computational study are promising drug lead molecules which can be considered for detailed experimental validation against RVFV infection.

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Introduction

The zoonotic disease Rift Valley fever is mosquito-borne and is prevalent among humans and animals in sub-Saharan African countries [1, 2]. The name Rift Valley originates from Kenya's Great Rift Valley, where the causative virus for this disease was first identified [3, 4]. The Rift Valley fever virus (RVFV) belongs to the genus *Phlebovirus* and the family *Phenuiviridae* (previously known as *Bunyaviridae*) [5, 6]. RVFV is transmitted to humans via mosquito or by the contamination with infected animal body fluids or tissues [3]. The Rift Valley fever disease has symptoms such as photophobia, headaches, retinitis, encephalitis, hemorrhagic fever, blindness and neurological effects [2, 5]. Originally endemic to sub-Saharan African regions, RVFV has spread outside to the Arabian peninsula in 2000 [7,8,9]. The World Health Organization (WHO) has expressed the need to accelerate research and development towards effective

therapies and vaccines against RVFV, since the virus has potential to cause world-wide outbreak and public health emergency [10, 11]. During the COVID-19 pandemic, there are also reports of Rift Valley fever outbreak in Uganda [12], and more recently in Mauritania [13]. Although, the overall documented human mortality rate for RVFV infections is less than 1% [14], in the case of severe infection, the human mortality rate can steeply increase to 10 – 20% [15]. In addition, RVFV is potential bioterrorism agent, as the virus can spread easily through mosquitoes and livestock, and thus, necessitating studies towards development of safe therapies against the RVFV infection [16,17,18,19].

RVFV is a single-stranded tri-segmented negative-sense RNA virus [20]. The Large (L) and Medium (M) segments of the RVFV genome have negative polarities, while the Small (S) segment of the RVFV genome has ambisense polarity [20,21,22]. The M segment of the genome encodes for two glycoproteins (Gn and Gc), and a non-structural protein. The non-structural protein is expressed by itself (NSm1) or in fusion with the glycoprotein Gn (NSm2) [23, 24]. In the negative sense, the S segment of the genome expresses the nucleoprotein (N protein), and in the positive sense, the S segment expresses the non-structural protein NSs [21, 22]. L segment, the largest segment of RVFV genome, expresses L protein which plays an important role in the viral genome replication and mRNA transcription [25,26,27]. The L protein has an endonuclease domain at the N-terminal region, an RNA-dependent RNA polymerase in the middle and a cap-binding domain (CBD) at the C-terminal region. Generally, in Bunyaviruses such as RVFV, the L protein initiates the transcription process by a cap-snatching mechanism [26, 28, 29]. In the cap-snatching mechanism, both endonuclease and cap-binding domains of the L protein are actively involved. CBD recognizes the m⁷GTP cap-analogue present in the 5' region of the host cell mRNA. The endonuclease domain of the L protein cleaves off the 5' host mRNA cap, which are used as primers in the transcription of viral mRNA in the host cell [24, 28]. In short, the recognition of the host mRNA cap by the CBD of RVFV L protein is crucial for the initiation of viral transcription. Therefore, the CBD of RVFV L protein can serve as an effective drug target for viral inhibition.

Previously, multiple in vitro and computational (virtual) screening studies have been published in the literature with the aim of identifying potential inhibitors to combat RVFV infection or against RVFV proteins such as L protein, glycoprotein N (Gn), glycoprotein C (Gc), and nucleocapsid (N) proteins [19, 27, 30,31,32,33,34,35]. In particular, Fatima et al. [35], have performed virtual screening of more than 6000 natural products against multiple proteins of RVFV to identify potential natural product inhibitors. However, to date, there is no published study with the aim of identifying potential inhibitors for the CBD of RVFV L protein. In this contribution, we used established computational approaches to identify potential natural product inhibitors for the CBD of RVFV L protein.

Plants or plant-derived products continue to be valuable sources of therapeutics against viral infections [36,